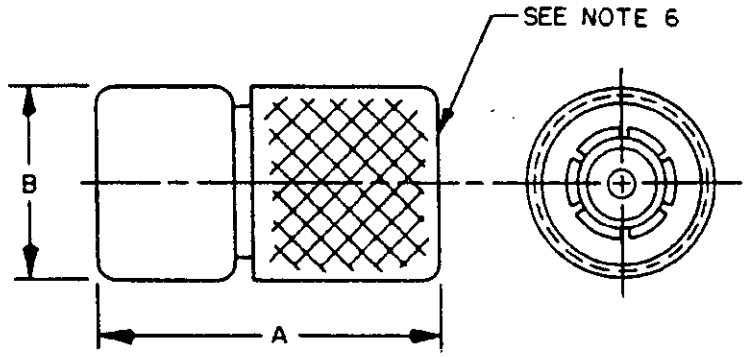
MILITARY SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY, (SERIES SC (CABLED), PIN CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-39012.

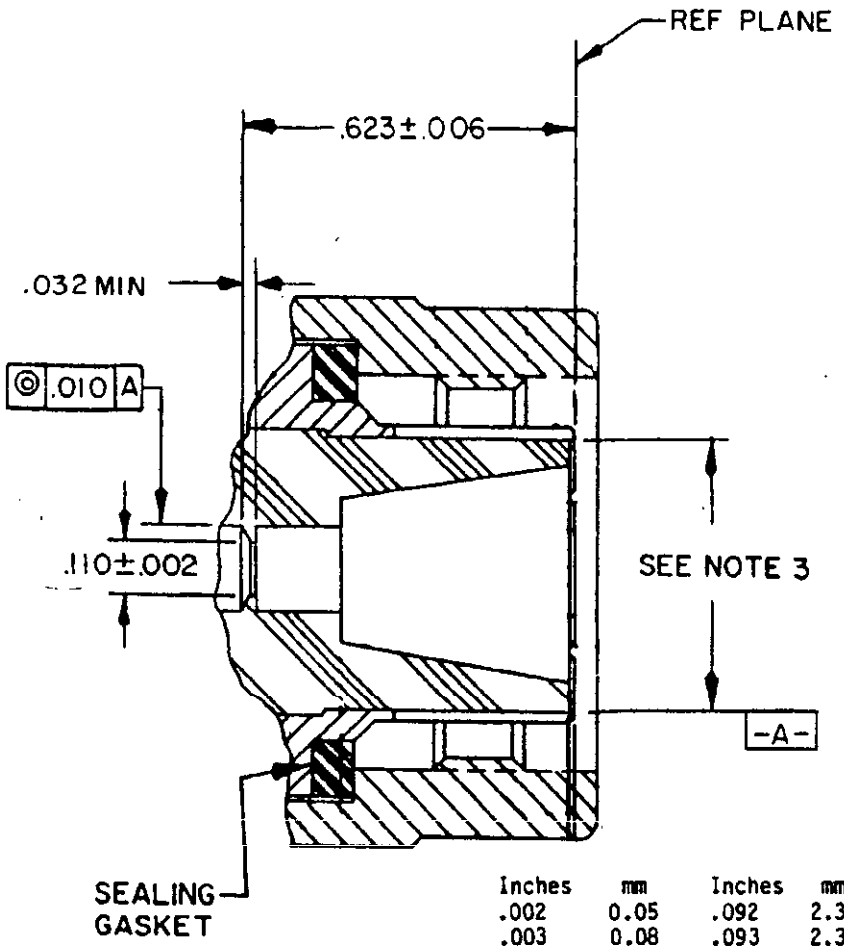
MARKING IMPLEMENTATION DATE FOR CATEGORY B, SEE TABLE IV



NOTES:
1. For dimensions A and B, see table I and III.
2. Dimensions B is the largest overall diameter of the connector.
3. Wrench flats are to accommodate standard wrench opening in accordance with H-28, appendix 10.
4. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
5. All undimensioned pictorial configurations are for reference purposes only.
6. Series SC, pin contact interface in accordance with MIL-STD-348, 309.1.

FIGURE 1. General configurations.

AMSC N/A
FSC 5935
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CATEGORY D

Inches	mm	Inches	mm	Inches	mm
.002	0.05	.092	2.34	.251	6.38
.003	0.08	.093	2.36	.276	7.01
.006	0.15	.110	2.79	.307	7.80
.007	0.18	.119	3.02	.309	7.85
.010	0.25	.124	3.15	.337	8.56
.025	0.64	.191	4.85	.623	15.82
.032	0.81	.194	4.93	.6875	17.46
.040	1.02	.213	5.41	.690	17.53
.085	2.16	.223	5.66	.828	21.03
.090	2.29	.250	6.35		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The ID of outer contact when inserted into a.411 (10.44 mm) maximum ring gage shall be .377 (9.58 mm) minimum.
4. Three holes .027 (0.69 mm) minimum diameter equally spaced for safety wiring. Location on coupling nut optional.
5. All undimensioned pictorial configurations are for reference purposes only.
6. Chamfer is optional.

FIGURE 2. Mating dimensions for pin contact terminations.

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TABLE I. Dash numbers, cross-reference, and dimensions.

Dash number 1/	Applicable cable 2/ M17/	Dimensions	Inches (millimeters) maximum
CATEGORY A - FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED)			
0016 3/	073-RG212 4/ 112-RG304	A	1.609 (40.87)
0017 3/	065-RG165 074-RG213 075-RG214 4/ RG-225/U 57	B	.828 (21.03)
0018 3/	078-RG217 4/	A B	1.875 (47.63) .891 (22.63)
0019 3/	072-RG211 4/	A	2.250 (57.15)
0020 3/	079-RG218 4/	B	1.328 (33.73)
0021 3/	074-RG215 6/	A B	2.500 (63.50) .891 (33.73)
0022 3/	92-RG115 5/	A B	1.609 (40.87) .828 (21.03)
CATEGORY C - FIELD REPLACEABLE (MIL-C-22520 CRIMP TOOL) 7/ SEE FOOTNOTE NEXT TO APPLICABLE CABLE FOR CRIMP DIE			
0010 3/	073-RG212 4/ 8/ 112-RG304 8/		
0011 3/	065-RG165 5/ 9/ 074-RG213 4/ 9/		
0012 3/	75-RG214 4/ 9/ RG-225/U 5/ 9/	A B	2.250 (57.15) .828 (21.03)
0013 3/	6-RG11 4/ 9/ 10/ 62-RG14 4 5/ 9/ 10/		
0014 3/	92-RG115 5/ 9/		
CATEGORY D - FIELD REPLACEABLE - DEFINED PIECE PARTS 7/ 11/ 12/			
0501 3/	75-RG214 4/ RG-225/U 5/ 127-RG393	A B	1.718 (43.656) .828 (21.03)
0502 3/	074-RG213 4/		
0503 3/	060-RG142 4/ 5/ 128-RG400	A B	1.625 (41.28) .828 (21.03)

1/ For cross-reference of dash number to superseded part number or type designation, see table IV.

2/ The latest version of each cable shall be applicable.

3/ These connectors have captivated center contacts.

4/ Cable to be used when performing tests requiring cable except as in 5/ and 10/.

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5/ Cable to be used for the +200°C temperature cycling tests.

6/ Armored cable.

7/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4.

8/ M22520/5-35 Closure A or M22520/5-55 Closure A

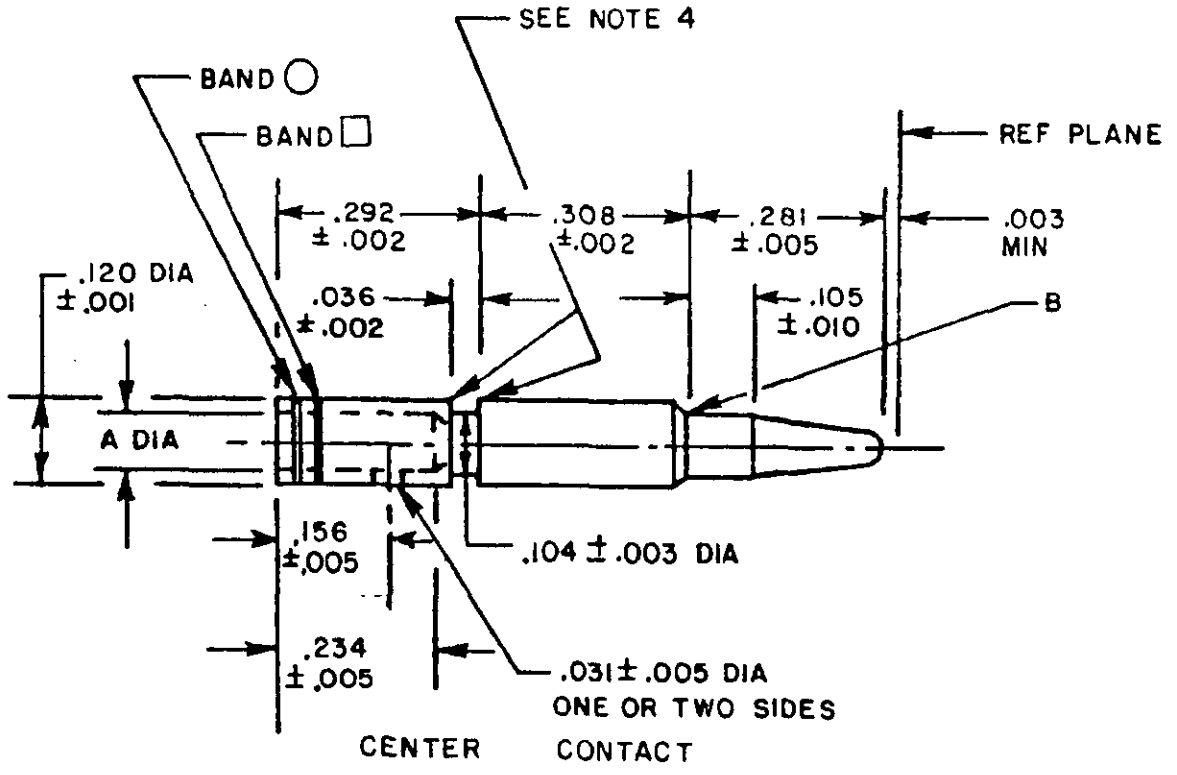
9/ M22520/5-61

10/ NOTE: These are not 50-ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

11/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

12/ Not to be used in Army equipment.

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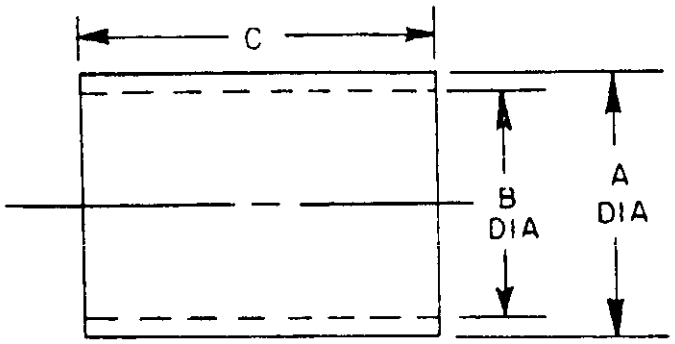


Dash no.	Contact no. 2/	A	B	Basic crimp tool 1/	Crimp die or positioner	Crimp tensile min	Color band □	Color band ○
0501 0502	35-10	.098 ±.002	.096 ±.002 X 45° ±5°	M22520/1-01	M22520/1-14	60 lbs	Red	Gray
0503	35-11	.043 +.001 -.002	.006 ±.002 X 30° ±5°	M22520/1-01	M22520/1-14	20 lbs	Blue	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Crimp tensile test shall be in accordance with MIL-C-39029.
4. .003 maximum break.
5. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.

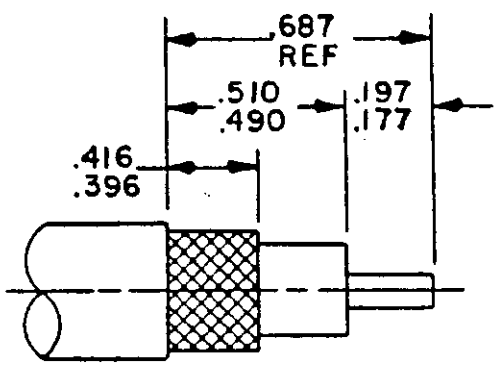
FIGURE 3. Contact and ferrule dimensions for category D only.

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Dash no.	Ferrule no. 2/	A	B	C ±.015	Basic crimp tool 1/	Crimp die or positioner M22520/5-
0501	35-50	.492 +.003 -.000	.438 +.003 -.000	.600		25 Closure A or 61
0502	35-51	.492 +.003 -.000	.418 +.003 -.000	.600	M22520/5-01	
0503	35-52	.250 ±.001	.220 ±.001	.500		5, 11, 57 Closure A or 19 Closure B

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.



Inches	mm
.177	4.50
.197	5.00
.396	10.06
.416	10.57
.490	12.45
.510	12.95
.687	17.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:
1,000 volts rms maximum working voltage at sea level.

250 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Design and configuration: See figure 1 and MIL-STD-348.

Force to engage and disengage:

Longitudinal force - Not applicable.

Torque - 6 inch-pounds maximum.

Coupling proof torque: 15 inch-pounds.

Inspection conditions: Coupling torque - 6 to 10 inch-pounds.

Mating characteristics: See figure 2 and MIL-STD-348 for dimensions.

Outer contact:

Test ring ID - .411 maximum, 16 microinch finish.

Insertion force - 15 pounds maximum when inserted a minimum of .125.

Contacts with slotted members: Shall contact a .419 minimum diameter ring within .031 of their tip ends.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B. 5,000 megohms minimum.

Center contact retention: (Applicable to captivated-center-contact/connectors only.) 15 pounds minimum axial force for all cables except RG400 and RG142; 6 pounds minimum for RG400 and RG142.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower; 1.30 maximum.

Swept frequency VSWR test setup:

Item 6 - VSWR shall be less than 1.015 +.005 F (F in GHz).

Item 16 - VSWR shall be less than 1.015 +.005 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.045 +.019 F (F in GHz).

Group B inspection - VSWR shall be less than 1.10 +.01 F (F in GHz).

Qualification and group C inspection - VSWR shall not exceed 1.15.

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Connector durability: 500 cycles minimum at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact	1.0	1.5
Outer contact	.15	Not applicable
Braid to body	.05	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 3,000 volts rms at sea level for connectors using other than RG400 and RG142; 1,500 Vrms minimum for connectors using RG400 and RG142.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see table I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage - 750 volts rms minimum.

Altitude - 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency - 2,500 volts rms tested at a frequency from 5 to 7.5 MHz.

Leakage current - Not applicable.

Cable retention force:

Noncrimp assemblies: 75 pounds minimum.

Crimp assemblies:

50 pounds minimum for cables .155-.189 OD.

60 pounds minimum for cables .190-.229 OD.

75 pounds minimum for cables .230-.249 OD.

90 pounds minimum for cables .250 OD and larger.

Coupling mechanism retention force: 100 pounds minimum.

RF leakage: -90 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss:
.15 dB maximum at 9 GHz.

.05 F (GHz) dB maximum tested at 3 GHz and 6 GHz.

Part number: M39012/35- (dash number from table I or "B" number from table III).

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TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
	M39012/35	M39012/35
I	-0017	-0015 -0016 -0017 -0018 -0019 -0020 -0021 -0022
II	B0024	B0023 B0024 B0025 B0026 B0027
III	-0011	-0010 -0011 -0012 -0013 -0014
IV	-0501 -0502	-0501 -0502 -0503
V	-0503	-0503

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

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TABLE III. CATEGORY B - NONFIELD REPLACEABLE (SPECIAL TOOLS MAY BE REQUIRED). 1/

[NOT FOR AIR FORCE OR NAVY USE. FOR OEM USE ONLY.]

Part no. 2/ M39012/35B	Applicable cable 3/ M17/	Dimensions	Inches (millimeters) maximum
0023 4/	073-RG212 5/ 112-RG304		
0024 4/	065-RG165 6/ 074-RG213 5/		
0025 4/	75-RG214 5/ RG-225/U 6/	A	2.250 (57.15)
		B	.828 (21.03)
0026 4/	6-RG11 5/ 7/ 62-RG144 6/ 7/		
0027 4/	92-RG115 6/		

1/ For maintenance replacements for category B, see table V.

2/ For cross-reference of dash number to superseded part number or type designation, see table IV.

3/ The latest version of each cable shall be applicable.

4/ These connectors have captivated center contacts.

5/ Cable to be used when performing tests requiring cable except as in 6/ and 7/.

6/ Cable to be used for the +200°C temperature cycling tests.

7/ NOTE: These are not 50-ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

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TABLE IV. Cross-reference of part numbers. 1/

Preferred part number M39012/35	Substitute for part number or type designation 2/
-0010	
-0011	
-0012	
-0013	
-0014	
-0016	M39012/35-0001
-0017	M39012/35-0002
-0018	M39012/35-0003
-0019	M39012/35-0004
-0020	M39012/35-0005
-0021	M39012/35-0015
-0022	
B0023	M39012/35-0006, M23329/5-01, M23329/5-03, M39012/35-0023
B0024	M39012/35-0007, M39012/35-0024
B0025	M39012/35-0008, M23329/5-02, M23329/5-04, M39012/35-0025
B0026	M39012/35-0009, M23329/5-09, M39012/35-0026
B0027	M39012/35-0027
-0501	
-0502	
-0503	

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The connectors that are in stock or distribution that were previously qualified and marked with the old part number shall also be considered acceptable for Government use until connector stock is purged.

2/ The superseded part number or the type designation is for cross-reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/35-XXXX shall be used in all cases for marking and identifying the connector.

TABLE V. Maintenance replacements for category B.

Category B number 1/	Category C dash number	Category A dash number	Category D dash number
B0023	0010	0016	---
B0024	0011	0017	0502
B0025	0012	0017	0501
B0026	0013	---	---
B0027	0014	0022	---

1/ Category B connectors are for original installation only. They will not be stocked or procured by the Government.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - EA, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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